

ABF-001 Reviewer Guide

Purpose

This guide directs independent reviewers to the load-bearing claims, evidence routes, and known limitations of ABF-001.

Suggested review sequence

1. Read the abstract, Sections 2-5, and the claim ledger.
2. Check the Reed-Muller degree threshold in Theorem 1.
3. Check the affine-coordinate spanning argument in Theorem 2.
4. Recompute the order-zero signature formula.
5. Run the primary verifier and six focused tests.
6. Run the independent bitset verifier.
7. Compare the two edge-atlas SHA-256 identities.
8. Inspect the corrected indexed-versus-distinct radical count.
9. Search for prior equivalent results under alternate terminology.

Principal claims to challenge

Claim	Failure mode to seek
Moment criterion	off-by-one degree threshold or incomplete monomial span
Bidual incidence	affine-coordinate convention error or nonlinearity in q
Punctured kernel	invalid parameter retained or valid parameter omitted
Symmetric pencil	matrix indexing error, asymmetry, or wrong kernel interpretation
511 exceptional hyperplanes	rank/nullity or invalid-parameter counting error
Rank histogram	truth-table or elimination error
469-edge double count	disagreement between mask-first and hyperplane-first routes
202 distinct subspaces	unrecognized additional duplicate or intersection
Forest census	missed cycle or component

Evidence hierarchy

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human proof
+ fresh NumPy implementation
+ separately written bitset implementation
+ historical separately written C implementation
+ exhaustive small-universe controls
+ deterministic hashes
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This hierarchy is substantial internal evidence. It is not external validation.

Corrected finite statement

The exact wording is:

The 255 nonzero masks contain 203 singular mask-indexed radicals representing 202 distinct nonzero radical subspaces. The only duplicate is the line 119d5, shared by masks 8a and 9b.

Do not shorten this to “203 distinct radicals.”

Review dispositions

Use one or more of:

- confirmed for stated scope;
- clarification required;
- proof gap;
- computational discrepancy;
- earlier equivalent theorem;
- known consequence;
- counterexample;
- scope reduction;
- suitable after revision;
- unsuitable for publication.

Claim boundaries

ABF-001 is not peer reviewed, proof-assistant verified, independently externally reproduced, historically first, or a full-width cryptographic result.